

Explainable AI

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Introduction

In this homework we implemented **LIME** and **Shapley values** techniques for explaining single instance predictions of black-box regression models. We used a provided toy dataset with 5000 instances, two continuous features $x_1, x_2 \in [-1, 0.5]$, one categorical feature $x_3 \in \{1, 2, 3\}$ and a continuous target y . The dataset exhibits a non-linear structure as shown in Figure 1. We trained several regressors, reported their performance on the test set and compared feature attributions across explanation methods on four synthetic examples [1, 2].

Methods

We used an 80/20 **train/test split** and trained three scikit-learn models: linear regression (LR), a multilayer perceptron (MLP) and gradient boosting (GB).

LIME

To explain a single black-box prediction, LIME first generates **1000 perturbed instances**: continuous features are sampled from normal distributions with means and standard deviations estimated from the training set, and categorical features are sampled according to their training frequencies. Each sample is predicted by the model and weighted by its distance to the original instance using an exponential kernel $w = \exp(-d^2/\sigma^2)$, where $\sigma = \sqrt{n} \cdot 0.75$ and n is the number of features. A sparse **Lasso surrogate** is then fitted on the perturbed instances, predictions and weights. Its coefficients are the LIME explanation.

Shapley Values

This technique samples **1000 background instances** from the training set and enumerates all $3! = 6$ feature orderings. For each ordering, feature values are replaced one by one from the sampled background values to the values of the explained instance and the change in model prediction $\hat{y}$ is measured as the marginal contribution. Contributions are **averaged across orderings and background samples** to obtain one Shapley value per feature and are explained relative to the expected prediction $\mathbb{E}[\hat{y}]$.

Results

The model results on 1000 test set instances are shown in Table 1. We observe that **complex models perform better** on this dataset, which motivates the use of black-box explanation techniques. Table 3 presents the four synthetic examples, model predictions and feature attributions for both methods. The comparison between LIME weights and Shapley values is also shown in Figure 2. Table 2 compares the LR coefficients with the local explanations.

Model	MAE	MSE	R ²
Gradient boosting	0.158	0.040	0.886
MLP	0.160	0.041	0.883
Linear regression	0.442	0.282	0.192

Table 1. Test set performance of the three evaluated models.

Example	Feature	LR Coef.	LIME	Shapley
1	$x_3 = 1$	+0.32	+0.48	+0.33
	x_1	-0.12	-0.12	-0.17
	x_2	+0.01	+0.01	-0.00
3	$x_3 = 2$	-0.15	-0.24	-0.14
	x_1	-0.12	-0.11	-0.17
	x_2	+0.01	-0.01	-0.00

Table 2. Linear regression coefficients, LIME weights and Shapley values for selected examples.

Discussion

We conclude that both methods are in **high agreement** across all examples, with the exception of Example 2, which lies in a **non-linear region** of the feature space. The LR coefficients are also surprisingly close to the local attributions from both methods. All three recognize x_1 and x_3 as the **important features**, while x_2 is **not**, which is also visible in the dataset visualization. These techniques can therefore be used after model training, both optimizing performance and keeping the possibility of explanation, even for complex models like neural networks.

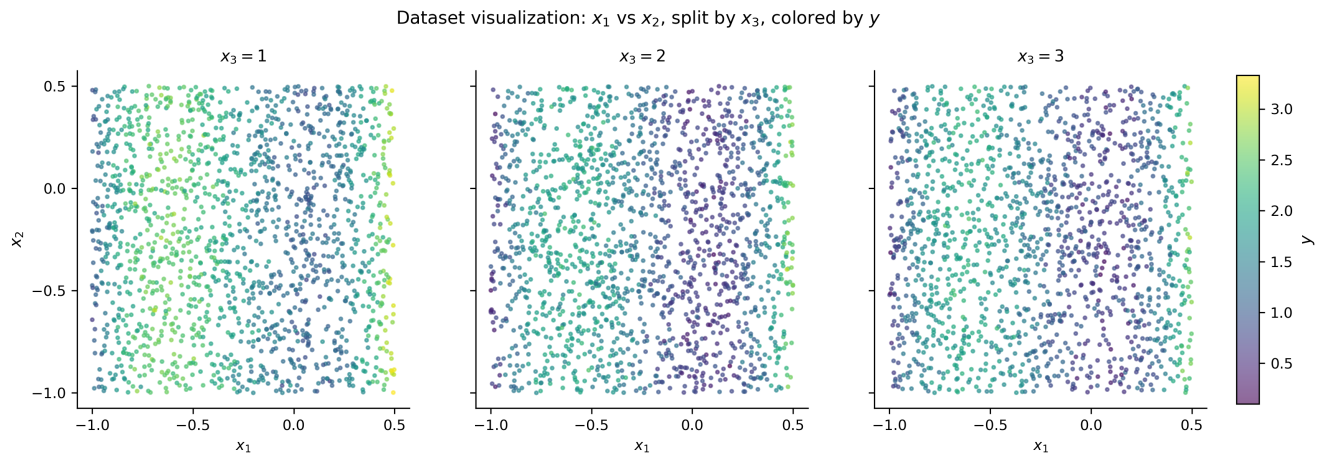


Figure 1. Toy dataset visualization showing the non-linear structure.

Example	x_1	x_2	x_3	Model	$\hat{y}$	w_{x_1}	w_{x_2}	w_{x_3}	ϕ_{x_1}	ϕ_{x_2}	ϕ_{x_3}
1	0.4	-0.4	1	GB	2.21	+0.45	+0.04	+0.41	+0.35	+0.01	+0.34
				MLP	2.22	+0.68	+0.05	+0.35	+0.39	-0.00	+0.31
				LR	1.66	-0.12	+0.01	+0.48	-0.17	-0.00	+0.33
2	0.2	-0.4	1	GB	1.30	+0.09	+0.03	+0.44	-0.58	+0.01	+0.35
				MLP	1.24	+0.20	+0.04	+0.40	-0.57	-0.02	+0.30
				LR	1.72	-0.12	+0.01	+0.48	-0.12	-0.00	+0.33
3	0.4	-0.4	2	GB	1.71	+0.44	+0.00	-0.26	+0.33	+0.01	-0.15
				MLP	1.79	+0.69	+0.01	-0.20	+0.40	+0.01	-0.14
				LR	1.19	-0.11	-0.01	-0.24	-0.17	-0.00	-0.14
4	0.4	+0.2	2	GB	1.70	+0.39	+0.01	-0.24	+0.33	+0.01	-0.16
				MLP	1.75	+0.58	+0.00	-0.23	+0.39	-0.00	-0.16
				LR	1.21	-0.11	+0.00	-0.22	-0.17	+0.01	-0.14

Table 3. Examples to explain, model predictions and comparison between LIME weights and Shapley values for each.

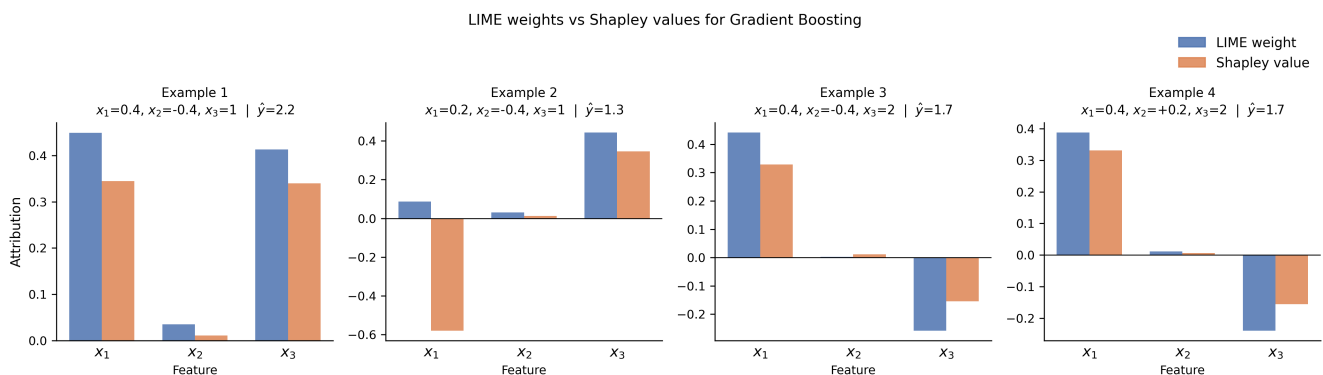


Figure 2. LIME weights and Shapley values comparison for GB predictions across all four examples.

References

- [1] Erik Štrumbelj and Igor Kononenko. A general method for visualizing and explaining black-box regression models. In *International conference on adaptive and natural computing algorithms*, pages 21–30. Springer, 2011.
- [2] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable

models instead. *Nature machine intelligence*, 1(5):206–215, 2019.